

1 Fundamentals of Classless Addressing (CIDR)

CIDR (Classless Inter-Domain Routing) removes traditional class boundaries and allows flexible allocation of IP address space.

An IPv4 address has **32 bits**:

[Network Bits | Host Bits]

If the prefix is **/25**:

- Network bits = 25
- Host bits = $32 - 25 = 7$

Address Calculations

- **Total IPs** = 2^H
- **Usable Hosts** = $2^H - 2$
(Network ID + Direct Broadcast Address reserved)

Example:

/25 → 7 host bits

$2^7 = 128$ total IPs

126 usable hosts

2 Subnetting Logic in CIDR

Subnetting = **borrowing host bits** to create smaller networks.

Each borrowed bit:

- Doubles number of subnets
- Reduces host capacity per subnet

Bits Borrowed	Subnets Created	New Prefix
1	2	+1
2	4	+2
3	8	+3

Example: Divide a /25 into 4 Equal Subnets

- Original host bits = 7
- Need 4 subnets $\rightarrow 2^2 = 4$
- Borrow 2 bits

New prefix = $25 + 2 = /27$

Remaining host bits = 5

$2^5 = 32$ IPs per subnet

30 usable hosts per subnet

3 Variable Length Subnet Masking (VLSM)

VLSM allows **different subnet sizes inside one block**, minimizing IP wastage.

Scenario:

From a /25 block (128 IPs), allocate:

- 60 hosts
- 30 hosts
- 30 hosts

Step 1: Allocate Largest Subnet First

60 hosts $\rightarrow$ needs 64 IPs

$2^6 = 64$

So we create a **/26** subnet.

Remaining space = 64 IPs.

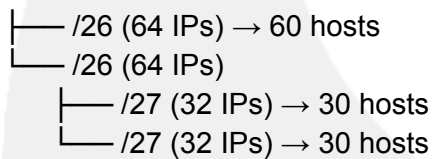
Step 2: Split Remaining Block

Each 30-host subnet needs 32 IPs (2^5).

So divide remaining /26 into two /27 subnets.

Final Allocation Structure

Original: /25 (128 IPs)



4 Bit Weightage (Critical for Fast Calculations)

In any octet:

Bit Position	Weight
7th	128
6th	64
5th	32
4th	16
3rd	8

2nd	4
1st	2
0th	1

Example:

If subnet mask in 4th octet = 224
 $224 = 128 + 64 + 32$

Block size = $256 - 224 = 32$

So network increments by 32:
0, 32, 64, 96, 128, 160, 192, 224

5 Block Validity Rule

For any block:

First IP must be divisible by block size.

Examples:

Block Size	Valid Starting Values
64	0, 64, 128, 192
32	0, 32, 64, 96, 128...
16	0, 16, 32, 48...

If block size = 64
192 is valid
130 is invalid

6 Point-to-Point Efficiency

Best prefix for 2 hosts = **/30**

Host bits = 2

$2^2 = 4$ IPs

Structure:

- Network ID
- Host 1
- Host 2
- Broadcast

Using /24 instead wastes 252 IPs.

7 Special IPv4 Address Rules

Loopback Addresses

Range: 127.0.0.0/8

- Used internally by host
- Never valid as source to external network

Limited Broadcast

255.255.255.255

- Used to broadcast to all devices on local network
- Cannot be routed

8 Fast Range Calculation Shortcut

Example:

$x.y.2.0 \rightarrow x.y.3.255$

Third octet changes across 2 values (2 and 3).

Fourth octet ranges 0–255 (256 values).

Total IPs = $2 \times 256 = 512$

$512 = 2^9$

Host bits = 9

Prefix = $32 - 9 = /23$

9 Finding the Last Assignable Host

Steps:

1. Identify Broadcast Address (all host bits = 1)
2. Subtract 1

Example:

Network = 192.168.1.0/26

Block size = 64

Broadcast = 192.168.1.63

Last usable = 192.168.1.62

10 Case Study: /20 Network Calculation

Given:

203.207.175.45/20

A /20 means:

- 16 bits fixed (first two octets)
- 4 bits fixed in third octet

Block size in third octet:

$256 - 240 = 16$

Valid third-octet network values:

0, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 224, 240

175 falls between 160 and 176

So network ID =
203.207.160.0/20

